



**Guidelines for framing
Curriculum and Credit Framework
for
Postgraduate Programmes in Biotechnology**

**W.E.F
AY-2026-27**

(Based on National Education Policy 2020)

Outline of the syllabus for M.Sc. in Biotechnology (Royal School of Bio-Sciences)

1 ST SEMESTER				
COURSE CODE	COURSE TITLE	LEVEL	CREDIT	L-T-P
BTC154C101	Biochemistry	400	3	3-0-0
BTC154C102	Microbiology	400	3	3-0-0
BTC154C103	Cell Biology	400	3	3-0-0
BTC154C104	Genetics	400	3	3-0-0
BTC154C115	Practical – I (A) *	400	4	0-0-8
BTC154C116	Practical – I (B) *	400	4	0-0-8
SWAYAM			3/ 4	
TOTAL CREDIT FOR 1 ST SEMESTER			20+3/4	
2 ND SEMESTER				
COURSE CODE	COURSE TITLE	LEVEL	CREDIT	L-T-P
BTC154C201	Molecular Biology	500	3	3-0-0
BTC154C202	Immunology	500	3	3-0-0
BTC154C203	Bioinformatics and Biostatistics	500	3	3-0-0
BTC154C204	Environmental Biotechnology	500	3	3-0-0
BTC154C215	Practical – II (A) #	500	4	0-0-8
BTC154C216	Practical – II (B) #	500	4	0-0-8
SWAYAM			3/4	
TOTAL CREDIT FOR 2 ND SEMESTER			20+3/4	
TOTAL CREDIT FOR 1 ST YEAR = 40 + 6/ 8				
3 RD SEMESTER				
COURSE CODE	COURSE TITLE	LEVEL	CREDIT	L-T-P
BTC154C301	Bioprocess Technology	500	4	4-0-0
BTC154C302	Analytical techniques	500	4	4-0-0
BTC154C303	IPR, Biosafety, Bioethics and Research Methodology	500	4	4-0-0
BTC154C304	Genetic Engineering	500	4	4-0-0
BTC154C315	Practical – III ^θ	500	4	0-0-8
TOTAL CREDIT 3 RD SEMESTER			20	
OR 3 RD SEMESTER (For students with 3 rd and 4 th Semester Research)				
BTC154R321	RESEARCH PROJECT – PHASE 1	500	20	
4 TH SEMESTER				
COURSE CODE	COURSE TITLE	LEVEL	CREDIT	L-T-P
BTC154C421	Dissertation (Students with research in 4 th Sem)	500	20	
(for 'Coursework only' students, in lieu of dissertation)				
BTC154C401	Plant and Animal Biotechnology	500	4	4-0-0
BTC154C402	Biophysical Chemistry	500	4	4-0-0
BTC154C403	Genomics and Proteomics	500	4	4-0-0
BTC154C404	Food Biotechnology	500	4	4-0-0
BTC154C415	Practical – IV [§]	500	4	0-0-8
TOTAL CREDIT 4 TH SEMESTER			20	
OR 4 TH SEMESTER (For students with 3 rd and 4 th Semester Research)				
BTC154R421	RESEARCH PROJECT – PHASE 2	500	20	
TOTAL CREDIT FOR 2 ND YEAR = 40				

Programme wise courses for each semester				
Programme	Semester	Mandatory Course codes	Level	Credits
Coursework only	1 st	All courses	400	20 + 3/ 4
	2 nd	All courses	500	20 + 3/ 4
	3 rd	BTC154C301, C302, C303, C304, C315	500	20
	4 th	BTC154C401, C402, C403, C404, C415	500	20
Coursework till 3 rd Semester and Research in the 4 th Semester	1 st	All courses	400	20 + 3/ 4
	2 nd	All courses	500	20 + 3/ 4
	3 rd	BTC154C301, C302, C303, C304, C315	500	20
	4 th	BTC154C421	500	20
Coursework (1 st and 2 nd semester) + Research only (3 rd and 4 th semester)	1 st	All courses	400	20 + 3/ 4
	2 nd	All courses	500	20 + 3/ 4
	3 rd	BTC154R321	500	20
	4 th	BTC154R421	500	20
Total Credits				80 + 6/ 8

** Practical – I (A) will include practical components from the theory papers ‘Biochemistry’ and ‘Microbiology’
Practical – I (B) will include practical components from the theory papers ‘Cell Biology’ and ‘Genetics’*

*# Practical – II (A) will include practical components from the theory papers ‘Molecular Biology’ and ‘Immunology’
Practical – II (B) will include practical components from the theory papers ‘Bioinformatics and Biostatistics’ and ‘Environmental Biotechnology’*

θ Practical – III will include practical components from the theory papers ‘Bioprocess Technology’, ‘Analytical Techniques’ and ‘Genetic Engineering’

§ Practical – IV will include practical components from the theory papers ‘Plant and Animal Biotechnology’, ‘Biophysical Chemistry’, ‘Genomics and Proteomics’ and ‘Food Biotechnology’

SYLLABUS (1st SEMESTER)

Course Title: Biochemistry	Course Component: Major
Course code: BTC154C101	Credit: 3
Level of course: 400	L-T-P-C: 3-0-0-3
Scheme of Evaluation: Theory (T)	

Course Objectives: The course is designed to understand the basic characteristics of various biological macromolecules, their formation along with their association in various metabolic pathways.

Course Outcome:

On successful completion of the course the students will be able to:		
Sl. No.	Course Outcome	Blooms Taxonomy Level
CO 1	Remember the core concept of basic biochemistry, structure of various biological macromolecules	BT 1
CO 2	Understand the basic biochemical processes occurring in the living system and involvement of various biological macromolecules in those processes.	BT 2
CO 3	Apply the knowledge gained during the course in the field of research and development.	BT 3
CO 4	Analyse theoretical knowledge in developing practical solutions in solving real life problems associated with biochemistry.	BT 4
CO 5	Evaluate their understanding in chemistry behind reactions occurring in living systems.	BT 5

Detailed Syllabus:

Modules	Topics / Course content	Periods
I	Chemical foundations of Biology: Composition of living matter, Water-properties, pH, pKa, acids, bases, buffers; weak bonds, covalent bonds. Protein: physical and chemical properties of amino acids; Primary, secondary, tertiary and quaternary structure; Globular and fibrous proteins; Amino acid composition and primary structure analysis, Structure-function relationship in model proteins like ribonuclease A, myoglobin and haemoglobin, structure of collagen, Ramachandran Plot	15
II	Carbohydrates: mono, di and polysaccharides; Structural and functional role; Glycoprotein and Glycolipid. Lipids: Structure and properties of storage and membrane lipids; Lipoproteins; Structural organization of biological membrane.	15
III	Nucleic acids: Structure and properties of purines, pyrimidines, nucleosides, nucleotides, helical structure of DNA. Different forms of DNA. Denaturation and renaturation of DNA. Enzyme catalysis: General principles of catalysis; Quantitation of enzyme activity and efficiency; Enzyme characterization and Michaelis-Menten kinetics; Relevance of enzymes in metabolic regulation, activation, inhibition and covalent modification; single and bi-substrate enzyme reactions.	15
IV	Metabolic pathways: Energy concepts and energy rich compounds; Glycolysis, glycogenolysis, gluconeogenesis, pentose phosphate pathway, citric acid cycle and oxidative phosphorylation; Fatty acid biosynthesis and oxidation (α and β) Vitamins: Types and biological properties	15
Total		60

Credit distribution:

3 credits: 3 × 30 = 90 Notional Credit Hours (60 class hours + 30 Experiential learning)

Experiential learning activities may include:

- undergraduate research
- participation in a student design team
- completion of an internship
- student teaching
- classroom presentation *etc.*

Textbooks

1. Nelson, D. L., Cox, M. M., *Lehninger Principles of Biochemistry*, 4th Edition, 2004, W.H. Freeman and Company, New York, USA
2. Satyanarayana, U. and Chakrapani, U, *Biochemistry*, 6th Edition, 2021, Elsevier.

Reference Books

1. Voet, D and Voet, J.G., *Biochemistry*, 4th Edition, 2012, Wiley
2. Berg, J.M., Tymoczko, J.L., Gatto, G.J. and Stryer, L, *Biochemistry*, 8th Edition, 2015, W.H. Freeman and Company

Course Title: Microbiology
Course code: BTC154C102
Level of course: 400
Scheme of Evaluation: Theory (T)

Course Component: Major
Credit: 3
L-T-P-C: 3-0-0-3

Course Objective: The course aims to give a holistic theoretical and practical knowledge in field of general microbiology, its core concept, scopes, applications and future prospects.

Course outcomes:

On successful completion of the course the students will be able to:		
SI No	Course Outcome	Blooms Taxonomy Level
CO 1	Remember the core concept of basic microbiology, microbial structure, their taxonomic classification, microbial ecology and their applications.	BT 1
CO 2	Understand isolation, screening, characterization, and identification of important microbes from various sources.	BT 2
CO 3	Apply the knowledge gained during the course in the field of research and development.	BT 3
CO 4	Analyse theoretical knowledge in developing practical solutions in solving real life problems associated with microbiology.	BT 4
CO 5	Evaluate future prospects by pursuing entrepreneurial ventures in this field.	BT 5

Detailed Syllabus:

Modules	Topics / Course content	Periods
I	Microbial Diversity and Systematics: Classical and modern methods and concepts	15

	in classification of microorganisms. Bergey's manual of determinative Bacteriology, 16s rDNA sequencing and ribosomal database project. Microbial systematics, Molecular Taxonomy,	
II	Study of microorganisms: General characteristics and salient features related to structure, function, physiology and significance of cyanobacteria, actinomycetes, fungi, yeast, viruses, rickettsia & mycoplasma. Ultrastructure of a bacterial cell: spore, cell wall, flagella, cell membrane, capsule, pili. Microbial growth. Virus structure and composition, virus replication and pathogenicity, Basic microbiological techniques: Microscopy, Pure culture, nutrition, enrichment, sterilization, disinfection, safety in the microbiological laboratory. microbial gene transfer: transformation, transduction, conjugation, plasmids, transposons.	15
III	Study of eco-physiological, biochemical and nutritional aspects of phylogenetically diverse representative groups of organisms: extremophiles - thermophiles, psychrophiles, halophiles, methanogens, archaeobacteria, Nitrogen fixing organisms and nitrogen fixing genes, Mycorrhiza: types and its functions Microbial Ecology: interactions among microbial populations, microbial interaction with animals, microbial interaction with plants, quorum sensing Diseases of humans: Bacterial meningitis, botulism, poliomyelitis, hepatitis and AIDS.	15
IV	Antibiotics: types & mode of action, resistance to antibiotics. Prebiotics and Probiotics, Bacteriocins, vaccines and adjuvants, Bioprocess technology, bioprocess control and monitoring variables, Media formulations, sterilization, Thermal death kinetics, batch and continuous sterilization systems, extracellular enzymes, biotechnologically important intracellular products, exopolymers.	15
Total		60
Pedagogy: Lectures, Assignments, Seminars		

Credit distribution:

3 credits: 3 × 30 = 90 Notional Credit Hours (60 class hours + 30 Experiential learning)

Experiential learning activities may include:

- undergraduate research
- participation in a student design team
- completion of an internship
- student teaching
- classroom presentation *etc.*

Textbooks

1. Willey, J., Sherwood, L., Woolverton, C.J., Prescotts Microbiology, , ISBN-10: 9813151269, ISBN-13: 978-9813151260, McGraw Hill Edition, 10th edition.
2. Ananthanarayan and Paniker's Textbook of Microbiology, ISBN-10: 9789386235251, ISBN-13: 978-9386235251, Universities Press
3. Reed, G., Prescotts and Dunn Industrial Microbiology, ISBN-10: 8123910010, ISBN-13: 978-8123910017, CBS Publishers & Distributors

Reference Books

1. Madigan, M.T., Martinko, J.M., Bender, K. S., Buckley, D.H., Stahl, D.A. Brock's Biology of Microorganisms, ISBN-10: 9332586861, ISBN-13: 978-9332586864, Pearson's Education,
2. Bauman R.W., Microbiology with Diseases by Taxonomy, Pearson Education, ISBN-10: 9332587272, ISBN-13: 978-9332587274

Course Title: Cell Biology
Course code: BTC154C103
Level of course: 400
Scheme of Evaluation: Theory (T)

Course Component: Major
Credit: 3
L-T-P-C: 3-0-0-3

Course Objective: The course aims to give a holistic theoretical and practical knowledge in field of cell biology, its core concept, scopes, applications and future prospects.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	Course Outcome	Blooms Taxonomy Level
CO 1	Ability to remember how cellular components are used to generate and utilize energy in cells.	BT 1
CO 2	Understand the structures and purposes of basic components of prokaryotic and eukaryotic cells, especially macromolecules, membranes, and organelles	BT 2
CO 3	Apply their knowledge of cell biology to selected examples of changes or losses in cell function. These can include responses to environmental or physiological changes, or alterations of cell function brought about by mutation.	BT 3
CO 4	Analyse the cell signalling and how it regulates cellular functions. Also, how its dysregulation leads to cancer and other diseases.	BT 4
CO 5	Evaluate the how cells grow, divide, and die and how these important processes are regulated.	BT 5

Detailed Syllabus:

Modules	Topics / Course content	Periods
I	Cell Structure and Methods in Cell Biology: Cell: Difference between prokaryotes and eukaryotes, structural and functional organization of eukaryotes, difference between plant and animal cells, Cell wall and cell membrane, Cell motility, sub cellular organelle like Nucleus, Endoplasmic reticulum, Golgi, Mitochondria, Lysosomes; Fractionation of sub cellular organelles, Principles and applications of the microscopy, Cell counting.	15
II	Bio-membrane structure and Function: Plasma Membrane: organization and properties, Dynamics transport across membrane, Cell signalling: Types of receptors (Intracellular and cell surface), signal transduction by membrane bound, cytosolic and nuclear receptors via various pathways Endo-membrane System and Cellular Motility: General organization of protein transport within and outside the cell, Mechanisms of endocytosis and exocytosis, Protein sorting and secretion, Vesicular transport, Mechanism of intracellular digestion.	15
III	Cell Dynamics: Cell dynamics, cytoskeleton and cell surface, Microfilaments: Structural organization, cell motility and cell shape; Microtubule: Structural and functional organization, cilia, flagella, centriole; Intermediate filaments, Cell-cell interactions and cell matrix interaction Cell Cycle & Cell Death: Mitosis, Meiosis, Eukaryotic Cell cycle and its regulation, Apoptosis, Cancer biology - Mechanism of carcinogenesis, tumor suppressor genes and oncogene.	15

IV	Cell Differentiation: Cell differentiation, hormones and growth factors; Stem cell differentiation, Blood cell formation, Fibroblast and their differentiation, Mating cell type in yeast, Surface antigen changes in Trypanosomes.	15
Total		60
Pedagogy: Lectures, Assignments, Seminars		

Credit distribution:

3 credits: 3 × 30 = 90 Notional Credit Hours (60 class hours + 30 Experiential learning)

Experiential learning activities may include:

- undergraduate research
- participation in a student design team
- completion of an internship
- student teaching
- classroom presentation *etc.*

Text Books

1. Gupta P.K., Genetics, ISBN-10 8171339328, ISBN-13 978-8171339328, Rastogi Publications, Meerut.
2. Watson, J. D., Baker T.A., Bell, S. P., Gann, A., Levine, M., and Losick, R. *Molecular Biology of the Gene*, 6th edition, 2008. Cold Spring Harbour Lab. Press, Pearson Pub.
3. De Robertis, E.D.P. and De Robertis, E.M.F. *Cell and Molecular Biology*, 8th edition, 2006, Lippincott Williams and Wilkins, Philadelphia.

Reference Books:

1. Karp, G..*Cell and Molecular Biology: Concepts and Experiments*, 6th edition, 2010. John Wiley & Sons. Inc.
2. Becker, W.M., Kleinsmith, L.J., Hardin. J. and Bertoni, G. P., *The World of the Cell*, 7th edition, 2009, Pearson Benjamin Cummings Publishing, San Francisco.

Course Title: Genetics	Course Component: Major
Course code: BTC154C104	Credit: 3
Level of course: 400	L-T-P-C: 3-0-0-3
Scheme of Evaluation: Theory (T)	

Course Objective: The course is designed to understand the various laws governing inheritance and learn about chromosomal aberrations and structure of chromosomes.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	Course Outcome	Blooms Taxonomy Level
CO 1	Remember the basic concept of Mendelian principles of heredity and use those principles to analyse genetic data.	BT 1
CO 2	Understanding of how genetic concepts affect broad societal issues including health and disease, food and natural resources, environmental sustainability, etc.	BT 2
CO 3	Apply to real life situations and one's life the principles of human heredity.	BT 3

CO 4	Analyse the historical and current knowledge regarding human heredity, and understand how such knowledge has influenced law, medicine, and society.	BT 4
CO 5	Evaluate the fundamentals of gene technology to understand how such technology impacts humans.	BT 5

Detailed Syllabus:

Modules	Topics / Course content	Periods
I	Mendelian genetics: Brief survey of Mendelian Genetics, law of dominance, independent assortment, linkage and crossing over, interaction of genes, Extrachromosomal inheritance: mitochondrial & chloroplast inheritance.	15
II	Microbial genetics: Bacterial chromosome and plasmids, bacterial mutants, prototroph and auxotroph. Transformation, conjugation and transduction in bacteria. Bacteriophage and their genetic systems, Lytic and Lysogenic cycles in lambda (λ) phage: genetic recombination and heteroduplex DNA.	15
III	Mutation: types, rates and the agents that cause mutation, Molecular basis of mutation, Genome instability: chromosomal aberration; Cell division and errors in cell division. Assay of mutagenic agents (Ames's test).	15
IV	Concept of Human Genetics: Human Chromosome and abnormalities, Mendelian pedigree pattern, polygenic and multifactorial inheritance, inborn errors of metabolism, Hardy-Weinberg equilibrium, genotype and allele frequency, sex determination, role of Y-chromosome and mechanism. Introduction to cancer genetics	15
Total		60
Pedagogy: Lectures, Assignments, Seminars		

Credit distribution:

3 credits: 3 × 30 = 90 Notional Credit Hours (60 class hours + 30 Experiential learning)

Experiential learning activities may include:

- undergraduate research
- participation in a student design team
- completion of an internship
- student teaching
- classroom presentation *etc.*

Textbooks

1. Gupta P.K., Genetics, ISBN-10 8171339328, ISBN-13 978-8171339328, Rastogi Publications, Meerut.
2. Watson, J. D., Baker T.A., Bell, S. P., Gann, A., Levine, M., and Losick, R., Molecular *Biology of the Gene*, 7th edition, 2012. Cold Spring Harbour Lab. Press, Pearson Pub.
3. Fairbanks, D. J., Genetics: The Continuity of Life, Wadsworth Publishing, ISBN-10: 0534252796
4. Russel, P. J., iGenetics, Pearsons Education India, ISBN-10: 9332571627, ISBN-13: 978-9332571624

Reference Books:

1. Karp, G., *Cell and Molecular Biology: Concepts and Experiments*, 6th edition, 2010. John Wiley & Sons. Inc.
2. Klug, W., Cummings, M., Spencer, C.A., Palladino, M.A., Concept of Genetics, ISBN-10: 9789332577466, ISBN-13: 978-9332577466, Pearsons Education India.

Course Title: Practical – I (A)
Course code: BTC154C115
Level of course: 400
Scheme of Evaluation: Practical (P)

Course Component: Major
Credit: 4
L-T-P-C: 0-0-8-4

Course Objective: The course is designed with an objective to give the students a wholesome practical knowledge on Microbiology, and Biochemistry.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	Course Outcome	Blooms Taxonomy Level
CO 1	Remember the practical skills associated with Microbiology, Genetics and Biochemistry.	BT 1
CO 2	Understand isolation, screening, characterization, and identification of important microbes from various sources.	BT 2
CO 3	Apply the knowledge gained during the course in the field of research and development.	BT 3
CO 4	Analyse theoretical knowledge in developing practical solutions in solving real life problems associated with microbiology.	BT 4
CO 5	Create an understanding in expanding the future prospects by pursuing entrepreneurial ventures in this field.	BT 5

Detailed Syllabus:

Modules	Topics / Course content	Periods
I	- Safety measures in Biochemistry laboratory. - Effect of α-amylase on starch - Determination of K_m and V_{max} of α-amylase activity - Preparation of phosphate buffer. - Preparation citrate buffer	24
II	- Principles of Colorimetry and Verification of Beer's law - Separation of Amino acids by paper chromatography. - Separation of various biomolecules by using TLC.	24
III	- Estimation of carbohydrates (glucose, maltose, lactose) present in a sample solution by DNS method - Estimation of protein by Lowry's method - Extraction and estimation of cellular protein from animal tissue by ammonium salt precipitation method.	24
IV	- Preparation of common bacteriological media and sterilization - Isolation and enumeration of microorganisms from various sources. - Staining of microorganisms (Bacteria and Fungi) - Biochemical characterization of microorganisms (IMViC test, catalase test, gelatine liquefaction, antibiotic sensitivity assay). - Growth curve, measure of bacterial population by standard plate count. - Antibiotic sensitivity assay	24
Total		96
Pedagogy: Lectures, Experiments, Laboratory sessions		

Credit distribution:

4 credits: $4 \times 30 = 120$ Notional Credit Hours (96 lab hours + 24 Experiential learning)

Experiential learning activities may include:

- undergraduate research
- participation in a student design team
- completion of an internship
- student teaching
- classroom presentation *etc.*

Texts and Reference: As suggested under theory papers.

Course Title: Practical – I (B) Course code: BTC154C116 Level of course: 400 Scheme of Evaluation: Practical (P)	Course Component: Major Credit: 4 L-T-P-C: 0-0-8-4
---	---

Course Objective: The course is designed with an objective to give the students a wholesome practical knowledge on Genetics and Cell Biology.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	Course Outcome	Blooms Taxonomy Level
CO 1	Remember the practical skills associated with Cell Biology and Genetics.	BT 1
CO 2	Understand the mechanism of cell division and differentiation.	BT 2
CO 3	Apply the knowledge identifying various mechanisms involved in cell growth and development	BT 3
CO 4	Analyze the equipment used and the underlying safety measures in a laboratory pertaining to cell biology.	BT 4
CO 5	Combine the concepts of cell biology and genetics to explain cellular anomalies related to genetic errors.	BT 5

Detailed Syllabus:

Modules	Topics / Course content	Periods
I	• Study the effect of temperature and organic solvents on semi permeable membrane. • Demonstration of dialysis. • Study of plasmolysis and de-plasmolysis. • Cell fractionation and determination of enzyme activity in organelles using sprouted seed or any other suitable source.	24
II	• Study of structure of any Prokaryotic and Eukaryotic cell. • Study of cell counting and their viability • Cell division in onion root tip/ insect gonads. • Identification of blood cells in human blood smear	24

III	- Calibration of Microscope - To prepare the media for culturing <i>Drosophila melanogaster</i> - To clean and sterilize the <i>Drosophila</i> culture bottle - Preparation of metaphase plate from mouse bone marrow. - Preparation of human karyotypes from well spread metaphase photographs	24
IV	- Temporary slide preparation of buccal smear to study Barr-body. - Introduction of chromosome abnormalities in mammalian chromosomes. - Temporary slide preparation of Grasshopper testis study the meiotic phases. - Temporary slide preparation of onion root tip to study the mitotic phases.	24
Total		96
Pedagogy: Lectures, Experiments, Laboratory sessions		

Credit distribution:

4 credits: $4 \times 30 = 120$ Notional Credit Hours (96 lab hours + 24 Experiential learning)

Experiential learning activities may include:

- undergraduate research
- participation in a student design team
- completion of an internship
- student teaching
- classroom presentation *etc.*

Texts and Reference: As suggested under theory papers.

SYLLABUS (2nd SEMESTER)

Course Title: Molecular Biology Course code: BTC154C201 Level of course: 500 Scheme of Evaluation: Theory (T)	Course Component: Major Credit: 3 L-T-P-C: 3-0-0-3
--	---

Course Objective: The course is designed to understand the organization of the prokaryotic/eukaryotic genome and the various molecular processes taking place in the living system

Course Outcome:

On successful completion of the course the students will be able to:		
SI No	Course Outcome	Blooms Taxonomy Level
CO 1	Remember the various molecular events associated with the growth and development of the cell.	BT 1
CO 2	Understand how replication, transcription and translation processes occur within the living cell.	BT 2
CO 3	Apply the knowledge gained during the course in the field of research and development.	BT 3
CO 4	Analyse the effects of various factors on molecular events including replication, transcription and translation.	BT 4
CO 5	Evaluate the knowledge to design experiments to manipulate cellular and molecular processes.	BT 5

Detailed Syllabus:

Modules	Topics / Course content	Periods
I	Genes and Chromosomes: Organization of bacterial genome; DNA structure, Structure of eukaryotic chromosomes; Complexity of genome and its reassociation kinetics (Cot curve analysis); Clusters and repeats; Chromatin: Heterochromatin and Euchromatin; Nucleosome structure and its phasing; DNase sensitivity, DNA methylation and imprinting, Human genome project and its importance, Structural genomics, Sequence components, Satellite, microsatellite and minisatellite chromosome.	15
II	Replication in prokaryotes & eukaryotes: Initiation and its regulation, elongation and termination in prokaryotes and eukaryotes; Enzymes and accessory proteins; Fidelity; Replication of single stranded circular DNA. Repair: Gene stability and Replication error repair, DNA repair enzymes: Photoreactivation, Nucleotide and base excision repair, Mismatch repair and SOS repair.	15
III	Prokaryotic & Eukaryotic Transcription: Promoters, Initiation, Elongation and Termination steps of prokaryotic transcription and its comparison with eukaryotic transcription. Enhancers, Transcription factors: TATA binding proteins (TBP) and TBP associated factors (TAF), Activators and repressors; Processing of primary transcripts; 5'-Cap formation; 3'-end processing and polyadenylation; Splicing; RNA editing; Translation & Transport: Translation machinery; Ribosomes; Steps of translation and its mechanism in prokaryotes and eukaryotes: Initiation, elongation and termination; Genetic codon and its properties; Co- and post translational modifications; Protein trafficking.	15
IV	Regulation of gene expression: Prokaryotic gene expression with reference to inducible and repressible operons, Concept of eukaryotic gene regulation, Chromatin remodelling, Epigenetics: Chromatin marking system; Regulatory RNA: Basic concepts of miRNA, siRNA and RNAi.	15
Total		60
Pedagogy: Lectures, Assignments, Seminars		

Credit distribution:

3 credits: 3 × 30 = 90 Notional Credit Hours (60 class hours + 30 Experiential learning)

Experiential learning activities may include:

- undergraduate research
- participation in a student design team
- completion of an internship
- student teaching
- classroom presentation *etc.*

Text Books

1. Watson, J. D., Baker, T. A., Bell, S. T., Gann, A. *Molecular Biology of the Gene*, Pearsons Education, 7th edition, ISBN 978-81-7758-181-2
2. Berk, A., Zipursky, S. L., Matsudaira, P.T., Baltimore, D., Darnell, J., Lodhish, H. F. *Molecular Cell Biology*, W.H. Freeman & Co Ltd (Latest Edition), ISBN-10: 0716731363, ISBN-13: 978-0716731368

Reference Books

1. Karp, G. *Cell and Molecular Biology: Concepts and Experiments*, 6th edition, 2010. John Wiley & Sons.

Inc.

2. Becker, W.M., Kleinsmith, L.J., Hardin. J. and Bertoni, G. P. *The World of the Cell*, 7th edition, 2009, Pearson Benjamin Cummings Publishing, San Francisco.

Course Title: Immunology	Course Component: Major
Course code: BTC154C202	Credit: 3
Level of course: 500	L-T-P-C: 3-0-0-3
Scheme of Evaluation: Theory (T)	

Course Objective: The course aims to give detailed concept in the core areas of immunology and understand the various forms of immunity and also the diseases associated with immune disorders.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	Course Outcome	Blooms Taxonomy Level
CO 1	Remember the basic forms of immune system present in the body.	BT 1
CO 2	Understand the mechanism of the immune system.	BT 2
CO 3	Apply the knowledge learnt in relating the same to the defence of the body during diseases.	BT 3
CO 4	Analyse the importance of the various molecules that play an important role in immune function.	BT 4
CO 5	Evaluate the various diseases that occurs in the system to the forms of immune disorders.	BT 5

Detailed Syllabus:

Modules	Topics / Course content	Periods
I	Immunology- fundamental concepts and anatomy of the immune system: Components of innate and acquired immunity; Phagocytosis; Complement and Inflammatory responses; Haematopoiesis; Organs and cells of the immune system- primary and secondary lymphoid organs; Lymphatic system; Lymphocyte circulation; Lymphocyte homing; Mucosal and Cutaneous associated Lymphoid tissue (MALT&CALT); Mucosal Immunity; Antigens - immunogens, haptens.	15
II	Immune responses generated by B and T lymphocytes: Immunoglobulins-basic structure, classes & subclasses of immunoglobulins, antigenic determinants; Multigene organization of immunoglobulin genes; B-cell receptor; Immunoglobulin superfamily; Antibody diversity, somatic hypermutation and class switching; Basis of self- and non-self-discrimination; MHC antigens and their role in immune responses, Kinetics of immune response, memory; B cell maturation, activation and differentiation; T-cell maturation, activation and differentiation and T-cell receptors; Functional T Cell Subsets; Cell-mediated immune responses, ADCC; Cytokines-properties, receptors and therapeutic uses; Antigen processing and presentation- endogenous antigens, exogenous antigens, non-peptide bacterial antigens and super-antigens; Cell- cell co-operation, Hapten-carrier system.	15

III	Antigen - Antibody Interactions: Precipitation, Agglutination; Advanced immunological techniques- RIA, ELISA, Western blotting, ELISPOT assay and Immunofluorescence. Complement system and complement fixation test. Clinical Immunology: Immunity to Infection: Bacteria, viral, fungal and parasitic infections (with examples from each group); Hypersensitivity - Type I-IV; Autoimmunity; Types of autoimmune diseases.	15
IV	Transplantation and tumor immunology: Transplantation- Immunological basis of graft rejection; Tumor immunology - Tumor antigens; Immune response to tumors and immune evasion by the tumor, Immunodeficiency-Primary and acquired immunodeficiency Vaccines: History, development, types and process of preparation and delivery	15
Total		60
Pedagogy: Lectures, Assignments, Seminars		

Credit distribution:

3 credits: 3 × 30 = 90 Notional Credit Hours (60 class hours + 30 Experiential learning)

Experiential learning activities may include:

- undergraduate research
- participation in a student design team
- completion of an internship
- student teaching
- classroom presentation *etc.*

Text books

1. Kuby, J., Thomas, J.K., Barbara, A.O. Immunology, 6th Edition, W. H. Freeman, 2002.
2. Janeway et al., Immunobiology, 4th Edition, Current Biology publications.,1999.

Reference books:

1. Brostoff, J., Seaddin, J.K., Male,D., Roitt, I.M., Clinical Immunology, 6thEdition, Gower Medical Publishing,2002.
2. Paul, R., Fundamental of Immunology, 4thedition, Lippencott,1999.
3. Goding, Monoclonal antibodies, Academic Press.1985.

Course Title: Bioinformatics and Biostatistics	Course Component: Major
Course code: BTC154C203	Credit: 3
Level of course: 500	L-T-P-C: 3-0-0-3
Scheme of Evaluation: Theory (T)	

Course Objectives: The course aims to give a holistic theoretical and practical knowledge in field of bioinformatics and biostatistics to understand the various cellular activities.

Course outcomes:

On successful completion of the course the students will be able to:		
SI No	Course Outcome	Blooms Taxonomy Level
CO 1	Remember the various software and biological databases and their application in the analysis of various biological experiments.	BT 1

CO 2	Understand the various software and their use in the analysis of various biological results.	BT 2
CO 3	Apply the knowledge to analyse the results of biological experiments statistically using various computational tools	BT 3
CO 4	Analyze the various biological events and their probable outcome using computational tools.	BT 4
CO 5	Evaluate various databases and software for the experiments/ analysis of the results of biological experiments.	BT 5

Detailed Syllabus:

Modules	Topics / Course content	Periods
I	Basics of bioinformatics: Definition, Scope and Goal, Application in Computational Biology, Limitations; Biological Database: Types of databases, biological database: GenBank, EMBL, DDBJ, Uniprot-KB: SWISS-PROT, PDB, Ace DBs, literature databases PubMed; Webtools: ExPASy server Sequence Analysis and Sequence Alignment: Basic concepts of sequence similarity, identity and homology, definitions of homologues, orthologues, paralogues and xenologues, Basic concepts of sequence alignment, Uses of Sequence Alignment, Pairwise, multiple, Database Similarity search, Scoring matrices: Basic concept of a scoring matrix, Matrices for nucleic acid and proteins sequences, PAM and BLOSUM series, matrix derivation methods and principles Sequence similarity search: BLAST and FASTA	15
II	Molecular Phylogenetics: Basic concepts, Methods in evaluation of phylogeny and steps in constructing alignments and phylogenetic Trees, Types of phylogenetic tree. Structural bioinformatics: proteins and its structure, Determination of protein 3Dstructure, Protein structure visualization, comparison, Secondary and tertiary structure prediction, Chemi-informatics and Computer Aided Drug Designing (CADD): Introduction to cheminformatics, Use of cheminformatics, Prospects of cheminformatics, Basics of medicinal chemistry. Prodrugs and soft drugs, Drug targets, Drug solubility, Natural resources of lead compounds, Pharmacokinetics & drug metabolism.	15
III	Statistical tools: Measures of central tendencies and dispersion, concept of probability and theoretical distributions (Binomial, Poisson and normal distribution), Correlation and Regression; Univariate and multivariate multiple regression. Random numbers, sampling methods, random plot design. Basics of testing of hypothesis. Analysis of variance (one way and two way), Students t test, Chi-square test, F-test and Z-test.	15
IV	Statistical Science and biological assays: Importance, nature and planning of bioassays; Direct and indirect bioassays; Design of experiments by Analysis of variance and Dose-response analysis. Analysis of biochemical data: Application of multiple regressions in epidemiologic and clinical data; Study of association between disease and risk factors. Application of odds ratio, Logistic regression with dichotomous response variable.	15
Total		60
Pedagogy: Lectures, Assignments, Seminars		

Credit distribution:

3 credits: 3 × 30 = 90 Notional Credit Hours (60 class hours + 30 Experiential learning)

Experiential learning activities may include:

- undergraduate research
- participation in a student design team
- completion of an internship
- student teaching
- classroom presentation *etc.*

Text Books

1. Zar, J. H. 2000. Biostatistical Analysis. Pearson Education, India.
2. Kothari, C. R. Research Methodology: methods and techniques. New Age International Publishers, India.

Reference Books

1. Quinn, G. P. & Keough, M. J. 2002. Experimental design and data analysis for biologists. Cambridge University Press, UK.
2. Gould. 2002. BioStats Basics. W H Freeman & Co, USA.

Course Title: Environmental Biotechnology	Course Component: Major
Course code: BTC154C204	Credit: 3
Level of course: 500	L-T-P-C: 3-0-0-3
Scheme of Evaluation: Theory (T)	

Course Objective: This course is offered with the objective of familiarizing students with the current and pertinent environmental issues and possible approaches to mitigate them.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	Course Outcome	Blooms Taxonomy Level
CO 1	Remember and identify area and time-specific environmental issues.	BT 1
CO 2	Understand the significance of environmental problems persisting in a place.	BT 2
CO 3	Apply the knowledge to relate cause and effect of major issues pertaining to the environment.	BT 3
CO 4	Analyze the scientific basis of the negative effects of pollutants on the environment.	BT 4
CO 5	Evaluate a detailed information system, starting from cause, effect, and solution to better prepare oneself to mitigate environmental concerns.	BT 5

Detailed Syllabus:

Modules	Topics / Course content	Periods
I	Environmental Pollution: Concept of Environmental Pollution; Origin of pollution; Classification and nature of Environmental Pollutants; Major sources; Impacts of Environmental Pollution at local regional and global level.	15
II	Air pollution: Concept of air Pollution; Major air pollutants and their sources; Meteorological aspects of air pollution; Oxides of nitrogen and sulphur; Particulate matter; Air pollution standards; Indoor and outdoor air pollution; Air pollution episodes and disasters; Effects of air pollution on human health, animals, plants,	15

	material and climate; Formation of fog and photochemical smog and acid rain; Monitoring of air pollution; Control on release of smoke.	
III	Soil Pollution: Concept of soil pollution; Causes of soil salinity; Different causes of soil degradation; Chemical and metallic pollution of agricultural soil; Mining and soil pollution; Control of soil pollution. Solid Waste: Concept of solid waste; Industrial solid waste; Domestic solid waste; Agricultural solid waste; Municipal solid waste; Major sources of solid wastes; Effects of solid waste generation on quality of air, water and public health; Technical approach for solid waste management; Disposal of organic and medical waste; Recovery and recycling of metallic waste; Disposal of plastic waste and hazardous wastes.	15
IV	Environmental Quality Assessment and Monitoring: What is environmental quality? Quality of environment for life on earth and man; Deterioration of environmental quality with reference to anthropogenic impact; Methods of assessment of environmental quality; Short term studies/surveys; Rapid assessment; Continuous short- and long-term monitoring Environmental Impact Assessment (EIA): Need of EIA; Scope and objectives; Types of environmental impacts; Steps involved in conducting the EIA Studies; Environmental Impact Assessment techniques; Merits and Demerits of EIA studies.	15
Total		60
Pedagogy: Lectures, Experiments, Laboratory sessions		

Credit distribution:

3 credits: 3 × 30 = 90 Notional Credit Hours (60 class hours + 30 Experiential learning)

Experiential learning activities may include:

- undergraduate research
- participation in a student design team
- completion of an internship
- student teaching
- classroom presentation *etc.*

Text Books

1. Wang, L. et al. (2010). Environmental Biotechnology, Humana Press. (available at UTS Library, either in hard copy or electronic version)
2. Wang, L. et al. (2010). Environmental Engineering, Humana Press. (available at UTS Library, either in hard copy or electronic version)

Reference Books

1. Vallero, D. A. (2010). Environmental Biotechnology: A Biosystems Approach, Elsevier. (available at UTS Library)
2. Evans, G. M. and Furlong, J. C. (2011). Environmental Biotechnology: Theory and Application, Wiley-Blackwell. (available at UTS Library)
3. Jördening, H. J. and Winter, J. (2005). Environmental Biotechnology: Concepts and Applications, Wiley-VCH. (available at UTS Library E-book)

Course Title: Practical – II (A)
Course code: BTC154C215
Level of course: 500
Scheme of Evaluation: Practical (P)

Course Component: Major
Credit: 4
L-T-P-C: 0-0-8-4

Course Objective: The course is designed with an objective to give the students a holistic practical knowledge on Molecular Biology and Immunology.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	Course Outcome	Blooms Taxonomy Level
CO 1	Remember the practical skills associated with molecular biology and immunology.	BT 1
CO 2	Understand the techniques of isolation of genetic material from different sources.	BT 2
CO 3	Apply the knowledge gained during the course in the field of research and development.	BT 3
CO 4	Analyse different DNA samples to establish identity of individuals.	BT 4
CO 5	Create an understanding in expanding the future prospects by pursuing entrepreneurial ventures in this field.	BT 5

Detailed Syllabus:

Modules	Topics / Course content	Periods
I	- Isolation of genomic DNA from plants/bacteria/animal cell. - Quantification and purity determination of isolated genomic DNA by UV-spectrophotometry and agarose gel electrophoresis.	24
II	- Extraction of RNA - Isolation of plasmid DNA by alkaline lysis and phenol method. - Restriction digestion of DNA - Polymerase chain reaction of genetic DNA - Agarose gel electrophoresis	24
III	- Blood film preparation, staining and identification of blood cells. - Preparation of antigen. - Immunization of mice, serum collection and preservation. - Purification of IgG from serum.	24
IV	- SGOT – PT test; agglutination. - Immuno-electrophoresis, Immuno-peroxidase test; Immuno-fluorescence test, ELISA. - Isolation of lymphoid cells (mouse) from spleen. - Separation of mononuclear cells.	24
Total		96
Pedagogy: Experiments and Laboratory sessions		

Credit distribution:

4 credits: $4 \times 30 = 120$ Notional Credit Hours (96 lab hours + 24 Experiential learning)

Experiential learning activities may include:

- undergraduate research
- participation in a student design team
- completion of an internship
- student teaching
- classroom presentation *etc.*

Texts and Reference: As suggested under theory papers.

Course Title: Practical – II (B) Course code: BTC154C216 Level of course: 500 Scheme of Evaluation: Practical (P)	Course Component: Major Credit: 4 L-T-P-C: 0-0-8-4
--	---

Course Objective: The course is designed with an objective to give the students a holistic practical knowledge on Bioinformatics and Environmental Biotechnology.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	Course Outcome	Blooms Taxonomy Level
CO 1	Remember the different bioinformatics databases.	BT 1
CO 2	Understand the process of molecular identification of organisms using bioinformatics tools.	BT 2
CO 3	Apply the knowledge in the field of environmental safety	BT 3
CO 4	Analyse different soil samples for their suitability for sustainable agriculture.	BT 4
CO 5	Develop frugal tests for and methods for industrial use.	BT 5

Detailed Syllabus:

Modules	Topics / Course content	Periods
I	- Introduction to bioinformatics software - Database search and sequence download - Sequence retrieval and analysis - BLAST, FASTA: Search and analysis of data	24
II	- Sequence alignment: algorithms for global and local alignments, pairwise alignment and multiple sequence alignment - Phylogenetic analysis and tree building - Protein structure download and structural analysis - ADMET – drug properties analysis and toxicology study	24
III	- Determination of BOD in contaminated water - Determination of COD in contaminated water - Screening of faecal coliform in water samples.	24
IV	- Screening of hydrocarbon degrading microbes from different environmental sources. - Production of secondary metabolites from environmental isolates. - Determination of soil and water pH from different locations.	24
Total		96
Pedagogy: Experiments and Laboratory sessions		

Credit distribution:

4 credits: 4 × 30 = 120 Notional Credit Hours (96 lab hours + 24 Experiential learning)

Experiential learning activities may include:

- undergraduate research

- participation in a student design team
- completion of an internship
- student teaching
- classroom presentation *etc.*

Texts and Reference: As suggested under theory papers.

SYLLABUS (3rd SEMESTER)

Course Title: Bioprocess Technology	Course Component: Major
Course code: BTC154C301	Credit: 4
Level of course: 500	L-T-P-C: 4-0-0-4
Scheme of Evaluation: Theory (T)	

Course Objectives: The course aims to give a holistic theoretical and practical knowledge in field of bioprocess technology involving microbial cultures, its core concept, scopes, applications and future prospects.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	Course Outcome	Blooms Taxonomy Level
CO 1	Remember the core concept of modern biotechnology and its application in food, pharma- and petroleum industries.	BT 1
CO 2	Understand the production procedure of alcoholic beverages, antibiotics and drugs.	BT 2
CO 3	Apply the knowledge gained during the course in the field of research and development.	BT 3
CO 4	Analyze theoretical knowledge in developing practical solutions in solving real life problems associated with microbiology.	BT 4
CO 5	Evaluate their understanding in expanding their future prospects by pursuing entrepreneurial ventures in this field.	BT 5

Detailed Syllabus:

Modules	Topics / Course content	Periods
I	Basic principle of Biochemical engineering: Isolation screening and maintenance of industrially important microbes, microbial growth and death kinetics (particularly with reference to industrially useful microorganisms), strain improvement for increased yield and other desirable characteristics	16
II	Detailed study of the design and operation of different types of fermenters, Mode of fermentation processes: Bioreactor designs, types of fermentations and fermenters: Upstream processing: scale up and scale down process. Fermentation process kinetics: Reaction kinetics: effect of temperature on reaction rate, activated complexes, catalyzed reactions, thermal death of micro-organisms, enzyme inhibition, Fermentation kinetics: advantages and limitations, Downstream processing: Bio separation: drying, crystallization, storage and packaging, treatment of effluent and its disposal	16

III	Applications of enzymes in food processing: enzymatic bioconversions e.g. starch and sugar conversion processes, High-Fructose Corn Syrup, and their downstream processing, backing by amylases, deoxygenation and desugaring by glucoses oxidase, beer mashing and chill proofing, cheese making by proteases. Application of microbes in food process operations and production: Fermented foods microbes and their use in pickling, producing colours and flavours, and process of wastes-whey, molasses, starch substrates and other food wastes for bioconversion to useful products; Bacteriocins: production and applications.	16
IV	Biodegradation of xenobiotic compounds and toxic wastes, removal of spilled oil & grease deposits, Biosurfactants, Bioremediation of soil & water, solid waste & waste water treatment, use of microorganism for the production of energy: Biogas (production of methane and hydrogen), fuel alcohol production & hydrocarbon production	16
TOTAL		64
Pedagogy: Lectures, Assignments, Seminars		

Credit distribution:

4 credits: $4 \times 30 = 120$ Notional Credit Hours (64 class hours + 16 Seminar + 10 Assignments + 30 Experiential learning)

Experiential learning activities may include:

- undergraduate research
- participation in a student design team
- completion of an internship
- student teaching
- classroom presentation *etc.*

Text Books

1. Fermentation and Biochemical Engineering Handbook, Principles, Process Design, and Equipment; Edited by Henry C. Vogel; Noyes Publications, New Jersey, U.S.A. ISBN: 0-8 155-1407-7.
2. Biotechnology- Volume 3- Bioprocessing; VCH Verlagsgesellschaft mbH. Weinheim, ISBN 3-527-28313-7 (Weinheim); ISBN 1-56081-153-6 (New York).

Reference Books

1. Principles of Fermentation Technology, P. E. Stanbury, A. Whitaker and S.J. Hall, Butterworth Heinemann, ISBN: 07506 45016.
2. Practical Fermentation Technology, B. Mcneil and L. M. Harvey, John Wiley & Sons Ltd, The Atrium, Southern Gate, Chichester, ISBN 978-0470-014349

Course Title: Analytical Techniques	Course Component: Major
Course code: BTC154C302	Credit: 4
Level of course: 500	L-T-P-C: 4-0-0-4
Scheme of Evaluation: Theory (T)	

Course Objective: The course is designed with an objective to give students the technical know-how of the working of analytical equipment used in Biotechnology.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	Course Outcome	Blooms Taxonomy Level
CO 1	Remember the specific technique to be used for different analytical characterizations.	BT 1
CO 2	Understand the working principles of various equipment used in analysis.	BT 2
CO 3	Ability to apply the acquired knowledge to address research problems.	BT 3
CO 4	Ability to analyse the data generated by using sophisticated equipment.	BT 4
CO 5	Ability to evaluate alternative and better methods of sample analysis to reduce time and increase throughput.	BT 5

Detailed Syllabus:

Modules	Topics / Course content	Periods
I	Basic techniques: Buffer preparations; pH measurement; Cell disintegration; Dialysis and Ultra filtration. Spectroscopy: Principles and applications of UV-Visible, Fluorescence and Infrared spectroscopy. Chromatography: Principles and applications of Paper and Thin layer chromatography; Size exclusion, Ion exchange, Hydrophobic, Reverse phase and Affinity chromatography; HPLC and FPLC.	16
II	Electrophoresis: Theory and application of Polyacrylamide and Agarose gel electrophoresis; Different variants of polyacrylamide gel electrophoresis (PAGE) like native, SDS-PAGE, 2D-PAGE, Blotting Techniques: Southern, Western and Northern blotting, Immunoblotting, Immunoelectrophoresis, Immunofluorescence, ELISA.	16
III	Centrifugation: Sedimentation, Analytical ultra-centrifugation, Preparative ultra-centrifugation: zonal and equilibrium density gradient ultracentrifugation. Radioactivity: Concept of radioactivity; Radioactivity counting methods with principles of different types of counters; Autoradiography; Applications of radioactive tracers in biology.	16
IV	Microscopy: Principles and applications of Simple, Compound and Phase contrast microscope, Fluorescence microscope, confocal microscope, Electron microscopy: SEM & TEM, Cryo-Electron microscopy	16
Total		64
Pedagogy: Lectures, Assignments, Seminars		

Credit distribution:

4 credits: 4 × 30 = 120 Notional Credit Hours (64 class hours + 16 Seminar + 10 Assignments + 30 Experiential learning)

Experiential learning activities may include:

- undergraduate research
- participation in a student design team
- completion of an internship
- student teaching
- classroom presentation *etc.*

Text Books

1. Wilson, K., and Walker, J. *Principles and Techniques of Practical Biochemistry*, 5th edition, 2000.
2. Freifelder, D., *Physical Biochemistry, application to Biochemistry and Molecular Biology*, 2nd edition, 1982.

Reference Books:

1. Holme, D., and Peck, H. *Analytical Biochemistry*, 3rd edition, 1998,
2. Scope, R. K. *Protein Purification: Principles and Practice*, 3rd edition, 1993.

Course Title: IPR, Biosafety, Bioethics and Research Methodology	Course Component: Major
Course code: BTC154C303	Credit: 4
Level of course: 500	L-T-P-C: 4-0-0-4
Scheme of Evaluation: Theory (T)	

Course Objective: The course aims to introduce students to Intellectual Property Rights and apprise them of ethical issues in the biological sciences and the laws pertaining to these in both the global and national context and also to aware the students with ethical practices appropriate for various scientific disciplines at all times and to adopt safe working practices relevant to the different biotech industries & fields of research.

Course outcome:

On successful completion of the course the students will be able to:		
SI No	Course Outcome	Blooms Taxonomy Level
CO 1	Remember intellectual property laws/principles (including copyright, patents, designs and trademarks) to real problems and to analyse the social impacts of intellectual property law and policy.	BT 1
CO 2	Understand , recognize and distinguish an ethical issue from other issues	BT 2
CO 3	Apply the knowledge gained during the course in spreading IPR related awareness.	BT 3
CO 4	Analyse experimental results for their potential to file suitable IPR.	BT 4
CO 5	Evaluate their understanding in expanding their future prospects by pursuing entrepreneurial ventures in this field.	BT 5

Detailed Syllabus:

Module s	Topics / Course content	Periods
I	Concept of Property: Tangible and Intangible Property, Intellectual Property-Origin Development and Objectives, Classification of Intellectual Property-Patents, Copyright, Trademark, Industrial Design, Geographical Indications, Protection of Plant Varieties and Traditional Knowledge, Relevance of Intellectual Property Rights for Science and Technology; Patentability Criterion-Discovery and Invention, Patentable Subject Matters; Novelty, Utility (Industrial Applicability), Non-Obviousness (Inventive Step) and Written Description, Product Patents vis-à-vis Process Patents; Patentability of Biotechnology Inventions; Patent Laws in Indian and International Perspective; Indian Patent Act 1970 (Patent Amendment Acts-1999, 2002 and 2005); International Conventions relating to Intellectual Property; General Agreement on Trade and Tariff (GATT); Trade Related Aspects of Intellectual Property Rights (TRIPS)	12

II	Biosafety: Definition and requirement; Important symbols and their meaning, Biosafety in relation to human health, environment, transgenic research and applications; International Legal Instruments on Biosafety Cartagena Protocol on Biosafety, Nagoya Protocol Laws relating to Biosafety in India: The Biological Diversity Act, 2002, Biosafety procedures, rules and guidelines under Environment (Protection) Act 1986 and Rules 1989; Biosafety Regulation: Principles and Practices in Microbial and Biomedical Labs; Guidelines for research involving genes; Regulatory bodies at National and International level	17
III	Nature, Concept and Relevance of Bioethics; Basic Principles of Bioethics; Legal, Social and Economic Impacts of the Products and Techniques in Biotechnology; Bioethics in Plants, Animals and Microbial Genetic Engineering; Ethical issues in Healthcare; Biopiracy and Bioethics: Application of IPR regime to Biological Resources and Biopiracy, Access to Biological Resources, Benefit Sharing and Informed Consent	17
IV	Meaning of research problem, Sources of research problem, Criteria Characteristics of a good research problem, Errors in selecting a research problem, Scope and objectives of research problem. Approaches of investigation of solutions for research problem, data collection, analysis, interpretation, Necessary instrumentations. Effective literature study approaches, analysis Plagiarism, Research ethics. Effective technical writing, how to write report, Developing a Research Proposal, Format of research proposal, presentation and assessment by a review committee.	18
Total		64
Pedagogy: Lectures, Assignments, Seminars		

Credit distribution:

4 credits: 4 × 30 = 120 Notional Credit Hours (64 class hours + 16 Seminar + 10 Assignments + 30 Experiential learning)

Experiential learning activities may include:

- undergraduate research
- participation in a student design team
- completion of an internship
- student teaching
- classroom presentation *etc.*

Text books

1. Cornish, W. R., Intellectual Property (Latest Edition)
2. Intellectual Property Rights by Paul Goldstein
3. Intellectual Property Rights by K. R. G. Nair, Ashok Kumar, K. R. G. Nair
4. Kilner, John, et.al, eds., Cutting-Edge Bioethics. Eerdmans 2002.

Reference Books:

1. B.L. Wadera, Patents, Trademarks, Copyright, Designs and Geographical Indications
2. S. Ignacimuthu, Bioethics, Alpha Science International, Limited (2009)
3. Matthew Rimmer, Intellectual Property and Biotechnology: Biological Inventions (2008)
4. Arthur L. Caplan, Robert Arp, Contemporary Issues in Bioethics (2014)

Course Title: Genetic Engineering
Course code: BTC154C304
Level of course: 500
Scheme of Evaluation: Theory (T)

Course Component: Major
Credit: 4
L-T-P-C: 4-0-0-4

Course Objective: The course aims to give in depth knowledge in field of genes and genetic engineering, the mechanism of creation of recombinant products and the role of instrumentation and sequencing process in genetic engineering.

Course Outcome:

On successful completion of the course the students will be able to:		
SI No	Course Outcome	Blooms Taxonomy Level
CO 1	Remember the core concept of Genetic Engineering, DNA modifying enzymes and cloning vectors <i>etc.</i>	BT 1
CO 2	Understand the production procedure of recombinant products by molecular cloning.	BT 2
CO 3	Apply the knowledge gained during the course in the field of research and development.	BT 3
CO 4	Analyze theoretical knowledge in developing biotechnological solutions in solving various problems.	BT 4
CO 5	Evaluate their understanding in expanding their future prospects by pursuing entrepreneurial ventures in this field.	BT 5

Detailed Syllabus:

Modules	Topics / Course content	Periods
I	Basics Concepts: DNA modifying enzymes; Cohesive and blunt end ligation; Linkers; Adaptors Homo-polymeric tailing; Labelling of DNA: Nick translation, Random priming, Radioactive and non-radioactive probes, Hybridization techniques: Northern, Southern and Colony hybridization, Fluorescence in situ hybridization; Chromatin Immunoprecipitation; DNA-Protein Interactions-Electromobility shift assay; DNase I foot printing	16
II	Cloning Vectors: Plasmids; Bacteriophages; M13 mp vectors; pUC19 and Bluescript vectors, Phagemids; Lambda vectors; Insertion and Replacement vectors; Cosmids; Artificial chromosome vectors (YACs; BACs); Animal Virus derived vectors-SV-40; vaccinia/ bacculo & retroviral vectors; Expression vectors; pMal; GST; pET-based vectors; Protein purification; His-tag; GST-tag; MBP-tag etc.; Intein-based vectors; Inclusion bodies; Methodologies to reduce formation of inclusion bodies; Baculovirus and Pichia vectors system, Plant based vectors, Ti and Ri as vectors, Yeast vectors.	16
III	Cloning Methodologies: Insertion of Foreign DNA into Host Cells; Transformation; Construction of libraries; Isolation of mRNA and total RNA; cDNA and genomic libraries; cDNA and genomic cloning; Expression cloning; Jumping and hopping libraries; Southwestern and Far-western cloning; Protein-protein interactive cloning and Yeast two hybrid system; Phage display; Principles in maximizing gene expression	16
IV	Application and study of gene regulation: DNA transfection, reporter assay, expression strategies for heterologous genes in bacteria, mammalian cells and plants. Targeted gene replacement. Sequencing methods: Enzymatic DNA sequencing; Chemical sequencing of DNA; high throughput DNA sequencing.	16

Total	64
Pedagogy: Lectures, Assignments, Seminars	

Credit distribution:

4 credits: 4 × 30 = 120 Notional Credit Hours (64 class hours + 16 Seminar + 10 Assignments + 30 Experiential learning)

Experiential learning activities may include:

- undergraduate research
- participation in a student design team
- completion of an internship
- student teaching
- classroom presentation *etc.*

Text books

1. Primrose, S.B., Twyman, R.M., and Old, R.W. *Principles of Gene Manipulation*. 6th Edition, S.B. University Press, 2001.
2. Brown, T.A., *Genomes 3*, 3rd ed. Garland Science, 2006.

Reference books

1. Sambrook, J., and Russel, D.W., *Molecular Cloning: A Laboratory Manual*, Vols 1-3, CSHL, 2001.
2. Selected papers from scientific journals.
3. Technical Literature from Stratagene, Promega, Novagen, New England Biolab, etc.

Course Title: Practical – III Course code: BTC154C315 Level of course: 500 Scheme of Evaluation: Practical (P)	Course Component: Major Credit: 4 L-T-P-C: 0-0-8-4
---	---

Course Objective: The course is designed with an objective to train the students in the practical aspect of bioprocess technology, analytical techniques used in biotechnological research, and genetic engineering.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	Course Outcome	Blooms Taxonomy Level
CO 1	Remember the practical skills associated with Cell Biology and Genetics.	BT 1
CO 2	Understand the process of isolation of production of fruit-based beverages.	BT 2
CO 3	Apply the knowledge gained during the course in the development of synthetic seeds.	BT 3
CO 4	Analyse different DNA samples employing different genetic engineering tools.	BT 4
CO 5	Create an entrepreneurial venture in field of analytics.	BT 5

Detailed Syllabus:

Modules	Topics / Course content	Periods
---------	-------------------------	---------

I	• Designing of stirred tank bioreactor • Production of wine from fruit juice • Isolation of protoplast from plant • Fusion of protoplast • Preparation of synthetic seeds • MBRT Test	24
II	• Isolation of genomic DNA from plant • Isolation of genomic DNA from animal • Isolation of genomic DNA from bacteria • Isolation of plasmid DNA from bacteria	24
III	• Determination of purity of DNA by spectrometric analysis • Restriction digestion of genomic DNA • PCR amplification of genomic DNA • Agarose gel electrophoresis	24
IV	• Paper chromatography • Thin layer chromatography • Silica gel column chromatography • Agarose gel electrophoresis • SDS-PAGE	24
Total		96
Pedagogy: Lectures, Experiments, Laboratory sessions		

Credit distribution:

4 credits: $4 \times 30 = 120$ Notional Credit Hours (96 lab hours + 14 Assignments + 10 Experiential learning)

Experiential learning activities may include:

- undergraduate research
- participation in a student design team
- completion of an internship
- student teaching
- classroom presentation *etc.*

Texts and Reference: As suggested under theory papers.

OR

3rd SEMESTER (*For students with 3rd and 4th Semester Research*)

Course Title: Research Project – Phase 1	Course Component: Research
Course code: BTC154R321	Credit: 20
Level of course: 500	
Scheme of Evaluation: Project presentation and Viva voce	

SYLLABUS (4th SEMESTER)

Course Title: Dissertation – 1 **	Course Component: Research
Course code: BTC154C421	Credit: 20
Level of course: 500	
Scheme of Evaluation: Project presentation and Viva voce	

**** *Only for Students with research in the 4th Semester***

For 'Coursework only' students, in lieu of dissertation

Course Title: Plant and Animal Biotechnology	Course Component: Major
Course code: BTC154C401	Credit: 4
Level of course: 500	L-T-P-C: 4-0-0-4
Scheme of Evaluation: Theory (T)	

Course Objectives: The objective is to teach the students the structure and function of plants and animals so that they learn about the structure of organs, organ systems, and cells, and how they function.

Course Outcome:

On successful completion of the course the students will be able to:		
SI No	Course outcome	Blooms Taxonomy Level
C01	Remember about the process of tissue culture, callus culture, suspension cell culture of plant tissues and the basic theories of animal cell culture.	BT1
C02	Understand the processes and techniques of creation of tissue cultured plants and animal cells and new hybrids.	BT2
C03	Apply the knowledge of in conservation of plants and animals and production of high yield animals.	BT3
C04	Analyze the theoretical knowledge in the generation of new plants and animal cell lines for experimental purpose.	BT4
C05	Evaluate the role of biotechnology in ensuring sustainable agriculture.	BT5

Detailed syllabus:

Modules	Topics & Course Contents	Period
I	Plant tissue cultures: callus, meristem culture etc., secondary metabolites in plant tissue cultures; protoplast culture and somatic hybridization; haploid plants and soma-clonal variation. Methods for Plant Conservation: Germplasm conservation- in situ, ex situ conservations and in vitro conservation; cryopreservation, Transgenic crop plants - Review of transgenic plants (Bt-cotton and other Bt- plants, Golden rice etc.), development of pathogen resistant cultivars using resistant lines.	16
II	Genetic engineering of crop plants: Agrobacterium-mediated gene transfer, direct gene transfer; Biolistic gene transfer, alternative approaches of gene transfer – micro injection and electroporation. Bioreactors- Stirred tank, Bubble column, Air lift, Rotating drum and immobilized plant cell reactor.	16
III	Animal Cell Culture: History of animal cell culture; Basic requirement for animal cell culture; Cell culture media and reagents; Animal cell; Tissue and organ cultures; Primary culture and established cell lines; Basic technique of mammalian cell culture, cell synchronization, Scaling of animal cell culture, Transfection and transformation of cells, Application of animal cell culture for in vitro testing of drugs; Testing of toxicity for environmental pollutants in cell culture; Stem cells and their application, in-vitro fertilization; culture of embryos; cryopreservation of embryos; embryo transfer; embryo splitting.	16
IV	Animal health Biotechnology: History of development of vaccines; Introduction to the concept of vaccines; Conventional methods and Recombinant approaches to vaccine production; Hybridoma technology; Application of cell culture technology in vaccine production and pharmaceutical proteins. Introduction of DNA fingerprinting. Scope and application of DNA forensics: animal species identification; detection of adulteration of meat by DNA based	16

	techniques.	
	Total	64
Pedagogy: Lectures, Assignments, Seminars		

Credit distribution:

4 credits: 4 × 30= 120 Notional Credit Hours (64 class hours + 16 Seminar + 10 Assignments + 30 Experiential learning)

Experiential learning activities may include:

- Undergraduate research
- Participation in a student design team
- Completion of an internship
- Student teaching
- Classroom presentation *etc.*

Text Books

1. Satyanarayana, U. Biotechnology (Books and Allied (P) Ltd.2005).
2. Slater, A., Scott, N. W., Fowler, M. R Plant Biotechnology: The Genetic Manipulation of Plants (Oxford University Press, USA; 2NDedition, 2008).
3. Animal cell biotechnology Portner, 2nd edition, Humana Press,2007.
4. Pinkert, Transgenic animal technology, Academic Press, 2006.

Reference Books

1. Adrian, S. Plant Biotechnology: The Genetic Manipulation Of Plants (Oxford University Press,2008)
2. Bohnert, H. J. et al Bioengineering and Molecular Biology of Plant Pathways, Volume 1 (Elsevier, USA,2008)
3. Davey, M.R. Plant Cell Culture: Essential Methods (Wiley-Blackwell Publishing, 2010)
4. Gordon, Reproductive technologies in farm animals, CAB Intl,2005.
5. Ed. John R.W. Masters, Animal cell culture- Practical approach, 3rd edition, Oxford University Press, 2000.

Course Title: Biophysical Chemistry	Course Component: Major
Course code: BTC154C402	Credit: 4
Level of course: 500	L-T-P-C: 4-0-0-4
Scheme of Evaluation: Theory (T)	

Course Objective: The course aims to give a holistic theoretical and practical knowledge in field of basics of Biophysical Chemistry, its role in the life form, and techniques to understand various Biophysical phenomena in living system.

Course Outcome:

On successful completion of the course the students will be able to:		
SI No	Course Outcome	Blooms Taxonomy Level
CO 1	Remember the practical skills associated with Biophysical Chemistry.	BT 1
CO 2	Understand molecular events associated with protein chemistry and basic principles associated with various instruments and techniques.	BT 2
CO 3	Apply the knowledge gained during the course in the field of research and development.	BT 3

CO 4	Analyse theoretical knowledge in developing practical solutions in solving real life problems associated with biophysical chemistry.	BT 4
CO 5	Evaluate their understanding in expanding their future prospects by pursuing entrepreneurial ventures in this field.	BT 5

Detailed Syllabus:

Modules	Topics / Course content	Periods
I	Interaction in biological systems: Intra and inter molecular forces, electrostatic interactions, hydrogen bonding, van der Waal interactions, hydrophobic interactions, disulfide bond. Biophysics of Water: Physicochemical properties of water, Molecular structure, Nature of hydrophobic interactions, Water Structure. Bioenergetics: Concept of energy coupling in biological processors, Energy requirements in cell metabolism, structure and role of mitochondria, high energy phosphate bond, energy currency of cell, biological oxidation, Electron-transport chain, Oxidative Phosphorylation including chemiosmotic hypothesis.	16
II	Protein Structure: Conformational properties of polypeptide, Ramachandran plot. Primary and secondary structure of proteins; alpha helix, beta sheet and random coil Tertiary structure; concept of domain and fold, Quaternary structure; Oligomeric proteins and cooperativity, Metalloproteins, Structural features of membrane proteins, intrinsically disordered proteins. Protein purification techniques: Gel filtration assay. SDS-PAGE Vs Native PAGE, 2D Gel electrophoresis	16
III	Multiple equilibrium: Titration of proteins to evaluate total and net charge; Scatchard and hill plots; Protein stability, denaturation, unfolding equilibrium; Kinetics and thermodynamics of protein folding; Protein refolding and aggregation; Effect of solvent and temperatures on the protein stability and folding, Heat Shock Proteins (Hsp) and their role in protein folding, scrapie proteins, Differential scanning calorimetry.	16
IV	Methods for the structure analysis: Far-UV and near UV-Circular Dichroism (CD); Fluorescence, single molecule fluorescence spectroscopy, fluorescent probes; Hydrogen-Deuterium (H-D) exchange; Fourier-transform Infrared (FT-IR) spectroscopy; Mass spectrometry (ESI and MALDI-TOF); Nuclear magnetic resonance (NMR) spectroscopy; X-ray crystallography.	16
Total		64
Pedagogy: Lectures, Assignments, Seminars		

Credit distribution:

4 credits: 4 × 30= 120 Notional Credit Hours (64 class hours + 16 Seminar + 10 Assignments + 30 Experiential learning)

Experiential learning activities may include:

- Undergraduate research
- Participation in a student design team
- Completion of an internship
- Student teaching
- Classroom presentation *etc.*

Text Books:

1. Nelson, D.L., Cox, M.M. *Lehninger Principles of Biochemistry*, 4th Edition, 2004, W. H. Freeman and Co., New York, USA

Reference Books:

1. Berg, J. M., Tymoczko, J. L. and Stryer, L. *Biochemistry*, 6th Edition, 2006, W.H. Freeman and Co.
2. Buchanan, B., Gruissem, W. and Jones, R. *Biochemistry and Molecular Biology of Plants*, 2nd Edition, 2015, American Society of Plant Biologists, USA.

Course Title: Genomics and Proteomics	Course Component: Major
Course code: BTC154C403	Credit: 4
Level of course: 500	L-T-P-C: 4-0-0-4
Scheme of Evaluation: Theory (T)	

Course objectives: The course is designed to appraise the students to the vital concepts of technologies pertinent to Genomics and Proteomics, their applications and demonstrate skills to apply the knowledge in scientific queries.

Course outcomes:

On successful completion of the course the students will be able to:		
SI No	Course Outcome	Blooms Taxonomy Level
CO 1	Remember the various techniques involved in the study of genomics and proteomics.	BT 1
CO 2	Understand the basic principle of all the techniques associated with genomics and proteomics study.	BT 2
CO 3	Apply the knowledge in the study of genomics and proteomics of a cell under specific conditions.	BT 3
CO 4	Analyze the effect of various intrinsic and extrinsic factors in the genome and proteome of a cell under certain conditions	BT 4
CO 5	Evaluate better and alternative methods to analyse the sample in cost effective manner.	BT 5

Detailed syllabus:

Modules	Topics / Course content	Periods
I	Genomics: DNA markers - SNP; STR; QTLs, RFLP; RAPD, cDNA and genomic libraries, Physical mapping of DNA by building genomic libraries, Clone contigs, YAC, BAC and PAC, Functional Genomics, DNA microarray, Functional analysis by gene knockouts	16
II	PCR and Its Applications: Primer design; Fidelity of thermostable enzymes; DNA polymerases, Types of PCR - multiplex, nested, reverse transcriptase, real time PCR, touchdown PCR, hot start PCR, colony PCR, cloning of PCR products; Basic concepts of genome sequencing, Next generation sequencing strategies, brief study about 3 rd and 4 th generation of sequencing	16
III	Basics of proteomics: Protein folding and modification, Types of proteomics, Protein sequencing, Protein structure determinations and Structural proteomics, Proteomic interactions (Y2H approaches, Co-IP); Concepts of protein engineering.	16

IV	Proteomic technologies: Microarray technology; Analytical proteomics tools (1-D & 2-D gel electrophoresis); Chromatography, in gel digestion, Mass spectrometry and analysis (ESI, MALDI), LC/MS-MS; Peptide mass fingerprinting.	16
Total		64
Pedagogy: Lectures, Assignments, Seminars		

Credit distribution:

4 credits: 4 × 30= 120 Notional Credit Hours (64 class hours + 16 Seminar + 10 Assignments + 30 Experiential learning)

Experiential learning activities may include:

- Undergraduate research
- Participation in a student design team
- Completion of an internship
- Student teaching
- Classroom presentation *etc.*

Text Books:

1. Discovering Genomics, Proteomics and Bioinformatics, 2nd Edition. Campbell AM & Heyer LJ, Benjamin Cummings 2007; CSH Press, NY. ISBN-10: 8131715590
2. Principles of Proteomics. R.M Twyman (2004) (BIOS Scientific publishers). ISBN-10: 1859962734
3. Genome III – T.A. Brown Garland Science Publ. June 08, 2006. ISBN-10: 0815341385

Reference Books:

1. Principles of Gene Manipulation and Genomics- Primrose S & Twyman R, 7th Edition, Blackwell, 2006. ISBN-10: 1405135441
2. Principles of Genome Analysis and Genomics. Primrose SB & Twyman RM. 2007. Blackwell. ISBN-10: 1405101202
3. Introduction to Genomics. A.M Lesk, Oxford University press, 2007. ISBN-10: 0199557489
4. A Primer of Genome Science. Greg Gibson and Spencer V. Muse. 2nd ed. 2004. SINAUER Associates Inc. ISBN-10: 0878932364

Course Title: Food Biotechnology	Course Component: Major
Course code: BTC154C404	Credit: 4
Level of course: 500	L-T-P-C: 4-0-0-4
Scheme of Evaluation: Theory (T)	

Course Objective: The main objective of the course is to provide the graduates with the knowledge of biofertilizers and their applications in agriculture.

Course Outcome:

On successful completion of the course the students will be able to:		
SI No	Course Outcome	Blooms Taxonomy Level
CO 1	Remember the various sources for production of food	BT 1
CO 2	Understand the principles and applications of enzymes and molecules	BT 2
CO 3	Apply the knowledge in practical applications for preparation of food	BT 3

CO 4	Analyze the role of biotechnology in food production.	BT 4
CO 5	Develop sophisticated methods for efficient production of traditional food items integrating biotechnology.	BT 5

Detailed Syllabus:

Modules	Topics & Course Contents	Periods
I.	Role of microorganisms in food production, History of the use of microorganisms in food, microorganisms in dairy products, meat industry and wine industry. Factors affecting microbial growth, Cancer causing foods, Contaminants in food, enzymes in food industry.	16
II	Colouring agents in food industry, Flavouring agents in food industry, Role of anti-oxidants, emulsifying and stabilising agents and food preservatives in flavour, Role of Flavour Enhancers, Stabilizers and Sweeteners in food industry	16
III	Food spoilage-causes and prevention, Food poisoning- causes and prevention, Foodborne diseases, refrigerated foods, canned foods, Dry Foods, Fermented and Pickled Foods, Food Labels and Allergens. Methods of Food preservation and processing, Methods in Food packaging.	16
IV	Fermented foods and their role in human health, Macro and Micronutrients from foods, Regulators of food industry, FSSAI, FDA, MOFPI, Regulations in production of GM foods. Benefits and risks of GM foods	16
Total		64
Pedagogy: Lectures, Experiments, Laboratory sessions		

Credit distribution:

4 credits: 4 × 30= 120 Notional Credit Hours (64 class hours + 16 Seminar + 10 Assignments + 30 Experiential learning)

Experiential learning activities may include:

- undergraduate research
- participation in a student design team
- completion of an internship
- student teaching
- classroom presentation *etc.*

Text Books:

1. Ratledge, C. and Kristiansen, B. (Eds.) (2006) Basic Biotechnology. 3rd Edition. Cambridge University
2. Johnson-Green, P. (2002). Introduction to Food Biotechnology

Reference Books:

1. Adams, M.R., and Moss, M.O. (2000). Food Microbiology. Second Edition. The Royal Society of Chemistry, UK.
2. Wood, B.J.B. (Editor) (1998). Microbiology of Fermented Foods, 2-Volumes, Second Edition. Blackie Academic & Professional, London.

Course Title: Practical – IV	Course Component: Major
Course code: BTC154C415	Credit: 4
Level of course: 500	L-T-P-C: 0-0-8-4
Scheme of Evaluation: Practical (P)	

Course Objective: The course is designed with an objective to train the students in the practical aspect of plant

and animal physiology, genomics and proteomics, and food biotechnology.

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	Course Outcome	Blooms Taxonomy Level
CO 1	Remember the practical skills associated with Cell Biology and Genetics.	BT 1
CO 2	Understand isolation, screening, characterization, and identification of important microbes from various food sources.	BT 2
CO 3	Apply the knowledge gained during the course in the field of research and development.	BT 3
CO 4	Analyse the genetic basis of health issues and provide solutions based on genomics and proteomics.	BT 4
CO 5	Develop scientifically sound and optimized methods for food analyses.	BT 5

Detailed Syllabus:

Modules	Topics / Course content	Periods
I	- Study of frog/human blood film. - Finding the coagulation time, blood groups, RBC count, TLC, DLC - Haemolysis: effect of isotonic, hypotonic, & hypertonic solution on Erythrocyte. - Determination of Haemoglobin - Preparation of haemin crystals.	24
II	- Preparation of stained mounts of anatomy of monocot and dicot's root, stem & leaf. - Demonstration of plasmolysis by Tradescantia leaf peel. - Demonstration of opening & closing of stomata - Demonstration of guttation on leaf tips of grass - Separation of photosynthetic pigments by paper chromatography. - Demonstration of aerobic respiration. - Preparation of root nodules from a leguminous plant.	24
III	- Primer designing for PCR amplification of DNA. - Protein isolation plant and animal samples. - Separation of proteins by SDS-PAGE - Detection of proteins using western blotting	24
IV	- Isolation of pathogens from food samples. - Estimation of proteins and carbohydrate from given food sample. - Estimation of antioxidant properties of green leafy vegetables - Estimation of fatty acids from a lipid sample - Adulteration test of dairy products	24
Total		96
Pedagogy: Lectures, Experiments, Laboratory sessions		

Credit distribution:

4 credits: 4 × 30 = 120 Notional Credit Hours (96 lab hours + 14 Assignments + 10 Experiential learning)

Experiential learning activities may include:

- undergraduate research
- participation in a student design team
- completion of an internship
- student teaching
- classroom presentation *etc.*

Texts and Reference: As suggested under theory papers.

OR

4th SEMESTER (For students with 3rd and 4th Semester Research)

Course Name: Research Project – Phase 2

Course code: BTC154R421

Level of course: 500

Scheme of Evaluation: Project presentation and Viva voce

Course Component: **Research**

Credit: 20